MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6C4WA

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, high frequency, medium Mu

Outline --- 5-2 (EIA)

Base --- E7-1

Envelope --- T5-1/2

Cathode --- Coated unipotential

Base connections:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Pin No. | --- | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| Element | --- | a | int con | h | h | a | g | k |

ABSOLUTE-MAXIMUM RATINGS:

| Parameter: | Ef | Eb | Ec | Ehk | Rg | Ik | Ic | Pp | Fl | TE | Alt |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | V | Vdc | Vdc | v | Meg | mAdc | mAdc | W | MHz | °C | ft |
| Maximum: | 6.9 | 330 | 0 | 135 | 0.5 | 18 | 5.0 | 3.4 | 150 | 165 | (See note 1) |
| Minimum: | 5.7 | --- | -55 | --- | --- | --- | --- | --- | --- | --- | --- |
| TEST CONDITIONS: | 6.3 | 250 | -8.5 | 0 | --- | --- | --- | --- | --- | --- | --- |

GENERAL

Qualification - Required



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| METHOD | REQUIREMENT OR TEST | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 1 | | | | | | | |
| 1201 | Short and discontinuity detection | | 0.4 | II | --- | --- | --- | --- |
| 1256 | Electrode current (1) (anode) | | 0.4 | II | Ib | 6.5 | 14.5 | mAdc |
| 1256 | Electrode current (2) (anode) | Ec = -25 Vdc | 0.4 | II | Ib | --- | 20 | µAdc |
| 1266 | Total grid current | Rg = 0.5 Meg (see note 2) | 0.4 | II | Ic | 0 | -0.5 | µAdc |
| 1301 | Heater current | | 0.4 | II | If | 140 | 160 | mA |
| 1306 | Transconductance (1) | | 0.4 | II | Sm | 1,750 | 2,650 | µmhos |
| 1336 | Heater-cathode leakage | | 0.4 | II | Ihk | --- | 10 | µAdc |
| | Quality conformance inspection, part 2 | | | | | | | |
| 1211 | Insulation of electrodes | | 2.5 | I | --- | --- | --- | --- |
| 1031 | High-frequency vibration | Rp = 2,000 ohms | 6.5 | Code E | Ep | --- | 50 | mVac |
| 1236 | Power oscillation | F = 150 MHz; Eb = 300 Vdc; Rg = 8,500 ohms (see note 3) | --- | --- | Po | 1.8 | --- | W |
| 2201 | Noise and microphonics | Ecal = 25 mVac; Rk = 0; Rp = 10,000 ohms | 2.5 | I | EB | --- | 17 | vu |
| 1256 | Electrode current (3) (anode) | Ec = -18 Vdc | 2.5 | I | Ib | 5 | --- | µAdc |
| 1266 | Grid emission | Ef = 7.6 V; Ec = -25 Vdc; Rg = 0.5 Meg (see note 4) | 2.5 | I | Ic | 0 | -1.0 | µAdc |
| 1306 | Transconductance (2) | Ef = 5.7 V | 2.5 | I | ΔSm Ef | --- | 15 | % |
| 1306 | Transconductance (3) | Eb = 100 Vdc; Ec = 0 | 2.5 | I | Sm | 2,500 | 4,000 | µmhos |
| 1316 | Amplification factor | | --- | --- | Mu | 15.5 | 18.5 | --- |
| 1331 | Direct-interelectrode capacitance | No shield | 6.5 | Code E | Cgp | 1.2 | 2.0 | pF |
| | | | | | Cin | 1.35 | 2.25 | pF |
| | | | | | Cout | 0.98 | 1.62 | pF |
| 2041 | Shock | 450 G; Ehk = 100 Vdc (see note 5) | 6.5 | See note 6 | --- | --- | --- | --- |
| 1031 | Vibration-fatigue | | 6.5 | See note 6 | --- | --- | --- | --- |

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| METHOD | REQUIREMENT OR TEST | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 2 -Continued | | | | | | | |
| --- | Shock and vibration - fatigue end points: | | | | | | | |
| 1031 | High-frequency vibration | | --- | --- | Ep | --- | 100 | mVac |
| 1336 | Heater-cathode leakage | | --- | --- | Ihk | --- | 15 | µAdc |
| 1306 | Transconductance (1) of individual tubes | | --- | --- | Sm | 1,600 | --- | µmhos |
| 1266 | Total grid current | | --- | --- | Ic | 0 | -1.0 | µAdc |
| 1121 | Base strain | | --- | --- | --- | --- | --- | --- |
| 2126 | Envelope strain | | 2.5 | I | --- | --- | --- | --- |
| 1105 | Permanence of marking | | --- | --- | --- | --- | --- | --- |

| METHOD | REQUIREMENT OR TEST | CONDITIONS | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 3 | | | | | |
| 1506 | Heater-cycling life | Ef = 7.5 V; Ehk = 135 Vdc; Eb = Ec = 0 | --- | --- | --- | --- |
| --- | Heater-cycling life end point: | | | | | |
| 1336 | Heater-cathode leakage | | Ihk | --- | 20 | µAdc |
| 1516 | Stability life | Ec = 0; Ehk = 135 Vdc; Rg = 0.5 Meg; Rk = 470 ohms; TA = room | --- | --- | --- | --- |
| --- | Stability-life end point: | | | | | |
| 1306 | Change in transconductance (1) of individual tubes | | $\Delta Sm_t$ | --- | 10 | % |
| 1501 | Intermittent life | Eo = 0; Ehk = 135 Vdc; Rg = 0.5 Meg; Rk = 470 ohms; TA = room; TE = 165°C (min) (see notes 7 and 8) | --- | --- | --- | --- |
| --- | Intermittent-life end points (500 hours): | See note 10 | | | | |
| --- | Inoperatives | | --- | --- | --- | --- |
| 1266 | Total grid current | | Ic | 0 | -0.5 | µAdc |
| 1301 | Heater current | | If | 140 | 162 | mA |
| 1306 | Change in transconductance (1) of individual tubes | | $\Delta Sm_t$ | --- | 20 | % |
| 1306 | Transconductance (2) | | $\Delta Sm_{Ef}$ | --- | 15 | % |
| 1336 | Heater-cathode leakage | | Ihk | --- | 10 | µAdc |
| 1211 | Insulation of electrodes | | R | 60 | --- | Meg |
| 1306 | Transconductance (1), average change | | $Avg \Delta Sm_t$ | --- | 15 | % |
| --- | Total defectives | | --- | --- | --- | --- |

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| METHOD | REQUIREMENT OR TEST | CONDITIONS | SYMBOL | LIMITS | | UNIT |
|---|---|---|---|---|---|---|
| | | | | MIN | MAX | |
| | Quality conformance inspection, part 3 - Continued | | | | | |
| --- | Intermittent life end points (1,000 hours): | See note 10 | | | | |
| --- | Inoperatives | | --- | --- | --- | --- |
| 1266 | Total grid current | | Ic | 0 | -0.5 | μAdc |
| 1301 | Heater current | | If | 140 | 164 | mA |
| 1306 | Change in transconductance (1) of individual tubes | | $\Delta Sm_t$ | --- | 25 | % |
| 1306 | Tranconductance (2) | | $\Delta Sm_{Ef}$ | --- | 20 | % |
| 1336 | Heater-cathode leakage | | Ihk | --- | 10 | μAdc |
| 1211 | Insulation of electrodes | | R | 50 | --- | Meg |
| --- | Total defectives | | --- | --- | --- | --- |
| 1511 | Cathode interface life | | --- | --- | --- | --- |
| --- | Cathode-interface life end points | Ef = 5.7 V; Eb = 80 Vdc; Ec/Ib = 2.0 mAdc (see note 9) | R | --- | 50 | Ohms |

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
2. This test shall be performed at the conclusion of the holding period.
3. An anode current of 25 mAdc on a bogey tube shall be obtained by adjusting the coupling between the load and the tank circuit while the load is simultaneously tuned to yield maximum power output.
4. Prior to this test the tube shall be preheated for a minimum of 5 minutes at the conditions indicated below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

| Ef | Eb | Ec | Rg | Rk |
|---|---|---|---|---|
| V | Vdc | Vdc | Meg | Ohms |
| 7.5 | 250 | 0 | 0.5 | 470 |

5. A grid No. 1 resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is employed.
6. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 12-month rule shall apply. MIL-STD-105, sample size code letter E, shall apply.
7. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
8. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, due to failure of more than 1 tube, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed three.

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NOTES: - Continued

9. No other test shall be made from the start of the interface life test until after the measurements of the end-point characteristic following completion of the indicated minimum number of the life-test hours.

10. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples.

11. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR,
Air Force - 99
DLA - ES

User activities:
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Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3230)

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